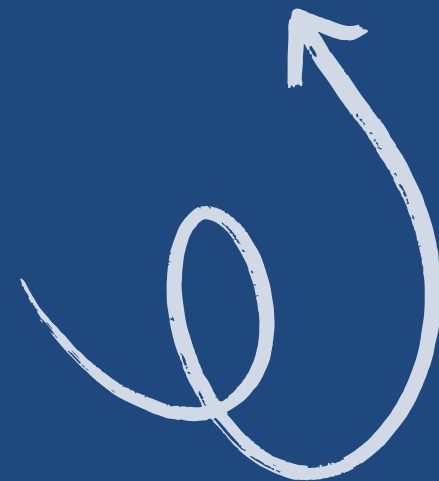


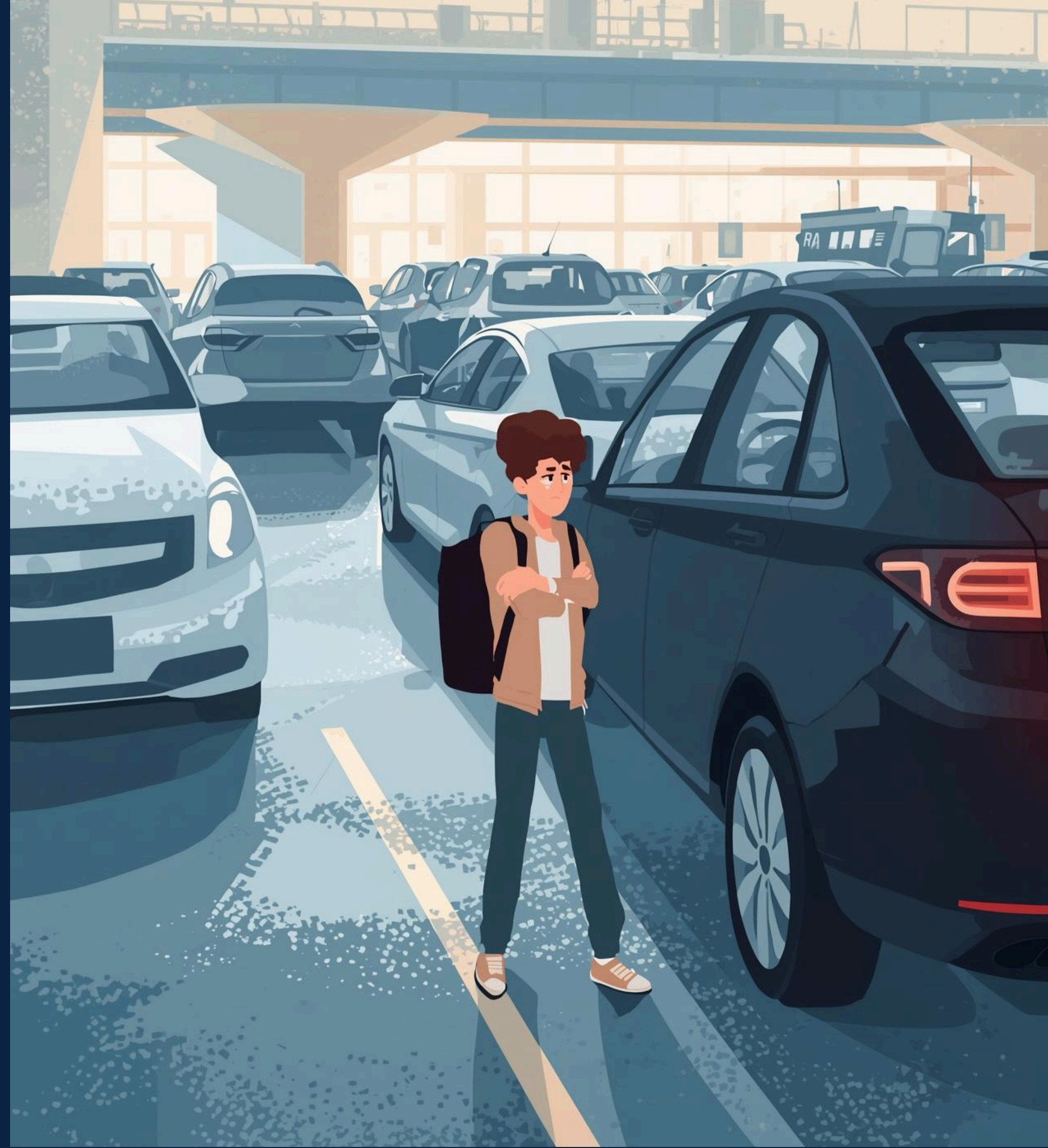
UTM Smart Parking Tracker

Presenter: Lim Yu Han



Parking Challenges for Students

Many students arrive early to campus yet find themselves late due to a lack of available parking. This frustrating experience leads to unnecessary stress and impacts their overall academic performance.



Current Parking System Challenges

The existing yearly sticker system creates **confusion and frustration**. Vehicles without stickers frequently enter restricted areas, leading to chaotic parking situations and a lack of enforcement exacerbates these issues.



Proposed Solution

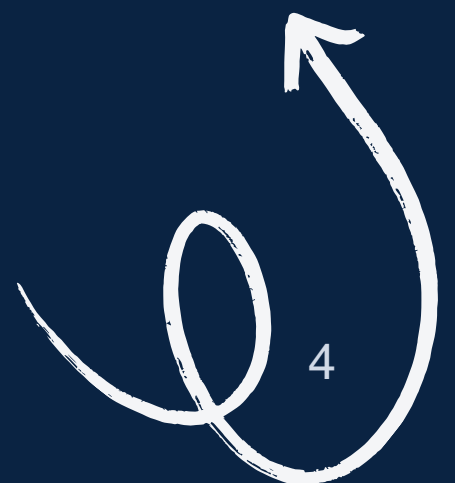
Introducing the UTM Smart Parking Tracker

Real-Time Tracking

The system provides **real-time availability updates**, allowing users to see open parking spots instantly, reducing time spent searching for parking and enhancing overall campus mobility.

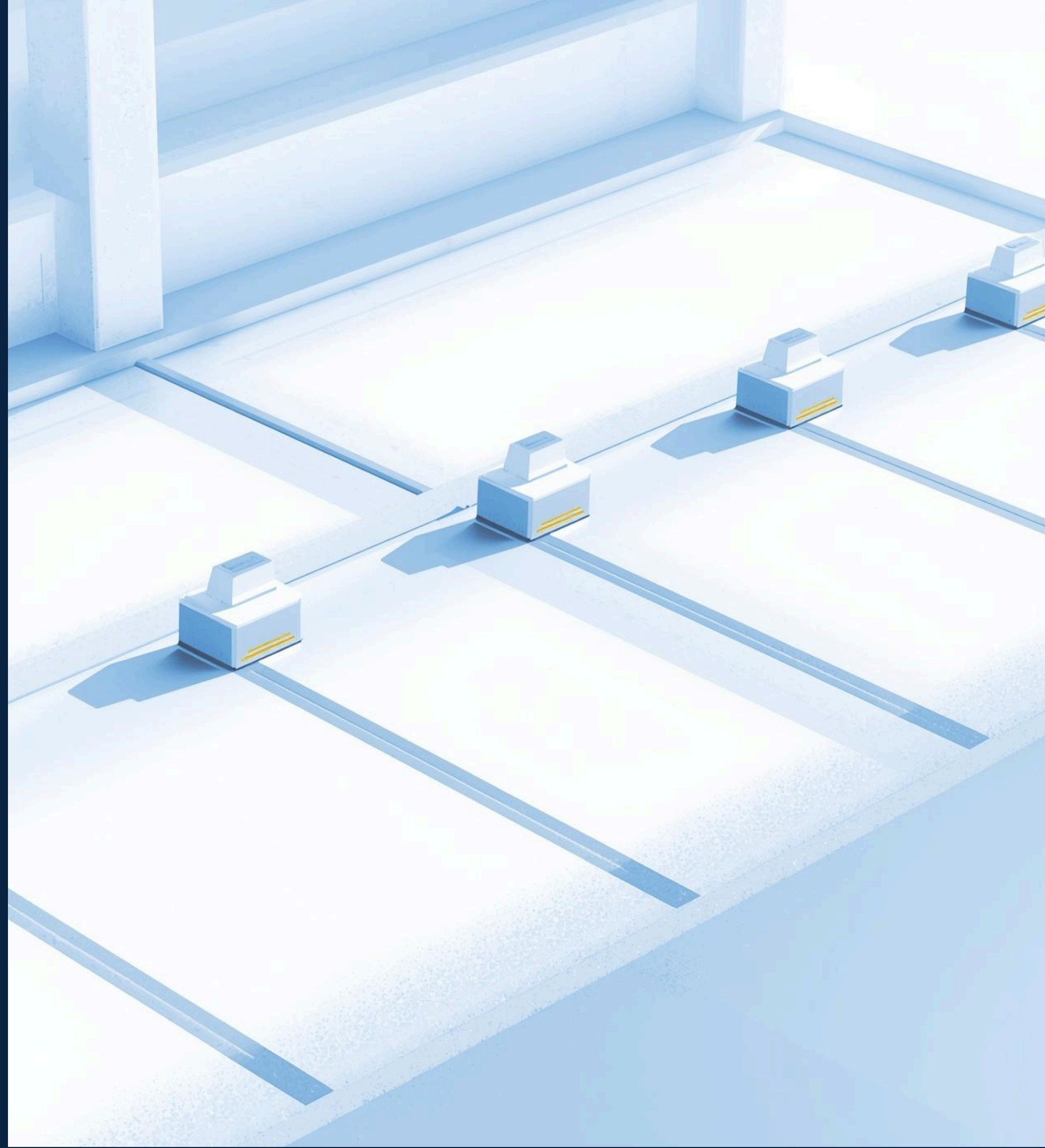
Automatic Verification

With automatic sticker verification, vehicles without valid stickers are flagged, ensuring compliance and improving parking management, which directly addresses current enforcement issues on campus.



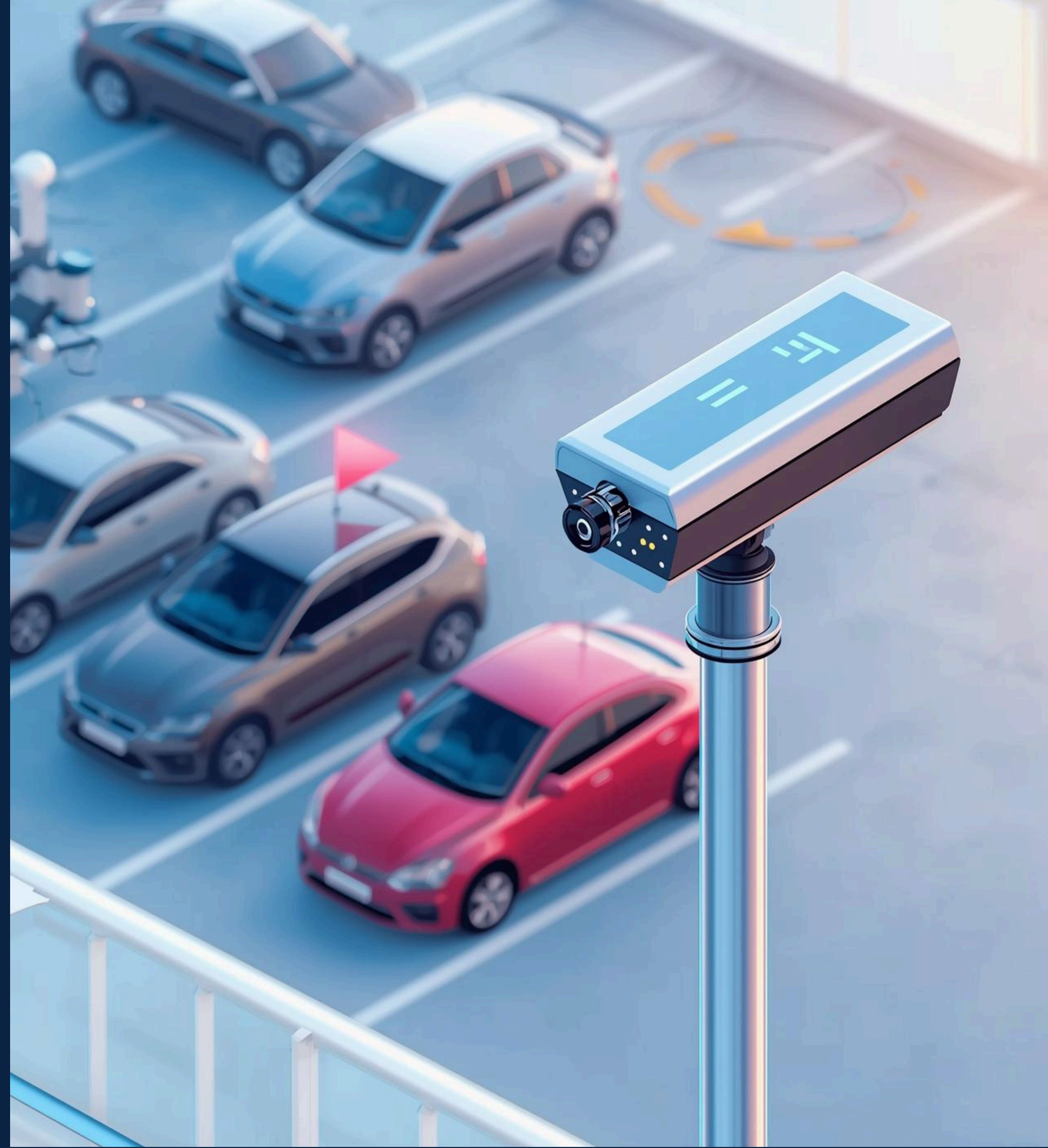
How IoT Sensors Enhance Parking

IoT sensors installed in each parking zone detect **actual slot occupancy**, providing real-time updates on parking availability. This data ensures drivers find open spots quickly and efficiently, reducing frustration and wait times.



AI Cameras for Parking Verification

The AI cameras will automatically verify sticker validity, flagging unverified vehicles. This system logs essential details like date, time, location, and plate number, enhancing enforcement efficiency on campus.



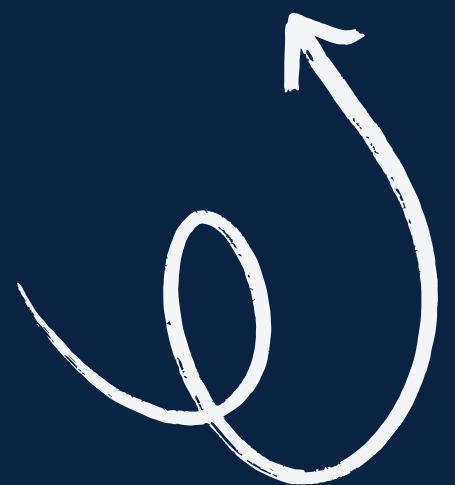
Proven Technology in Action

Many **Malaysian malls** have successfully implemented similar parking systems, showcasing reliability and efficiency. This technology adapts seamlessly to the UTM campus, ensuring a smart solution to our parking challenges.



Implementation and Budget Overview

The estimated budget for the UTM Smart Parking Tracker is **RM65,000**, covering essential components such as IoT sensors, AI cameras, software development, system integration, and a one-year maintenance plan.



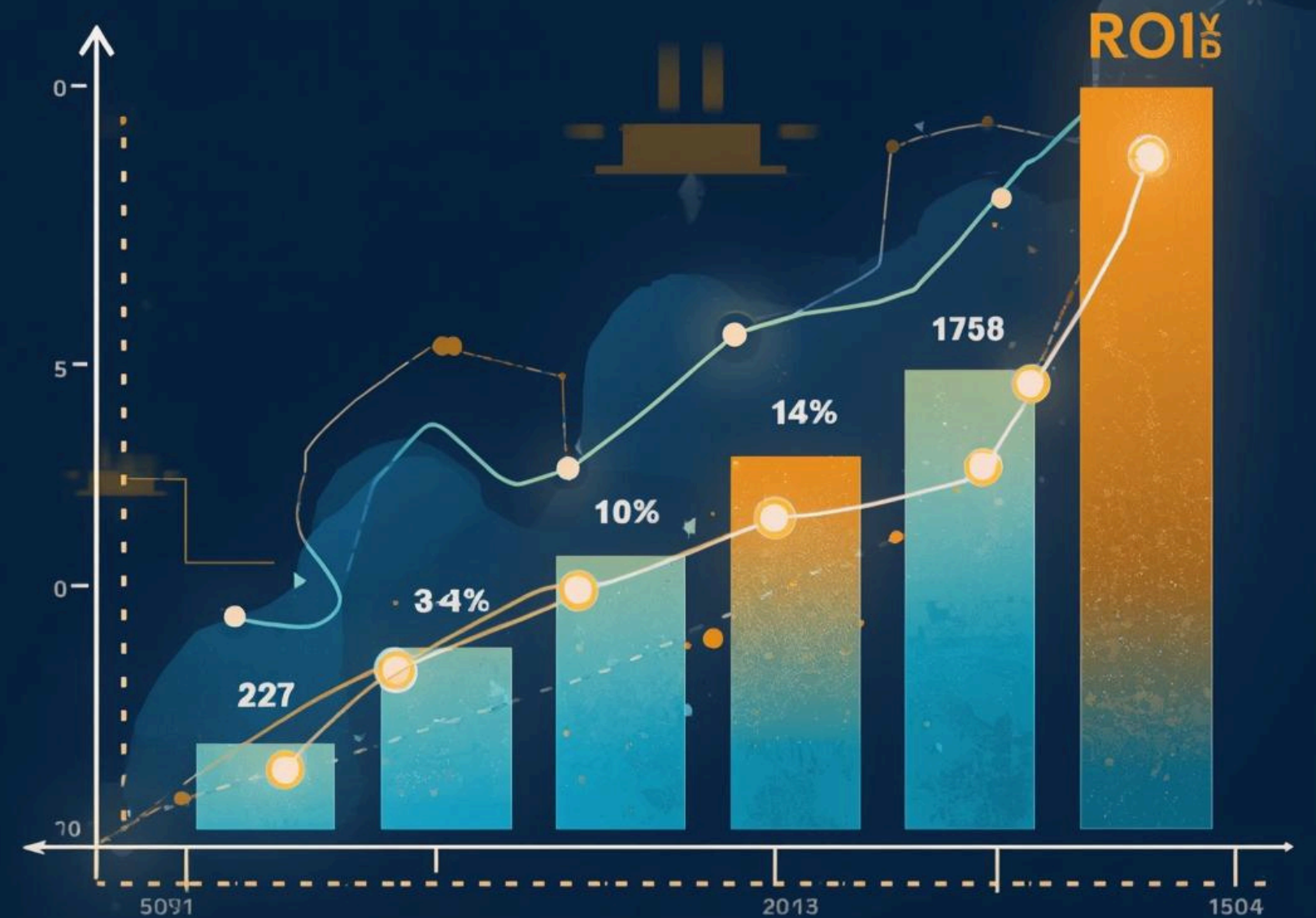
Enhanced Parking Experience for Students

The UTM Smart Parking Tracker allows students to **check parking availability** before leaving their accommodation, eliminating frustration and saving time, ensuring a smoother start to their campus day.



High ROI Potential for Investors

The UTM Smart Parking Tracker offers a **substantial return on investment**, with opportunities to license the system to other universities and generate long-term revenue through maintenance and AI monitoring subscriptions.





Conclusion

Transforming campus life for everyone